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## RESEARCH ARTICLE

# Design of an Explainable Deep Learning Framework for ECG-Based Arrhythmia Prediction

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**ABSTRACT** Arrhythmia refers to abnormal deviations in the normal heart rhythm, possibly signifying severe cardiovascular conditions. The manual identification of arrhythmias by ECG analysis is labor-intensive, requires specialized expertise, and is susceptible to inter-observer variability. Conventional signal-based classification techniques depend significantly on manually produced features, which may exhibit insufficient resilience and generalizability across varied patient demographics and noisy datasets. These constraints underscore the pressing need for automated, scalable, and interpretable diagnostic methodologies. To rectify these deficiencies, we offer an innovative approach that converts the ECG signal into 2D waveform images, facilitating the use of advanced image-based deep learning algorithms. ECG recordings from MIT-BIH were segmented around beats and converted to waveform graphs using wfdb, matplotlib, and PIL. Each image, reduced to  $224 \times 224$  pixels, graphically encapsulates about 0.7 seconds of ECG data, enabling deep neural networks to learn both temporal and morphological features. The preprocessing pipeline comprises amplitude normalization, noise filtration and augmentation to improve signal quality and model robustness. We introduce a hybrid deep learning model (ECG-CNNTX -Electrocardiogram Classification using CNN and Transformer with Explainability) that integrates EfficientNetB0 for effective and resilient feature extraction with a Transformer-based attention module to capture long-range relationships in image representations. Our model achieved a classification accuracy of 99% in identifying Premature Ventricular Contractions (PVCs). Grad-CAM visualization elucidates essential ECG areas that impact predictions, hence augmenting clinical dependability for model interpretability. This study introduces a dependable, precise, and interpretable ECG classification framework that integrates traditional signal processing with sophisticated AI-driven image analysis, providing an innovative and scalable method for arrhythmia diagnosis.

**INDEX TERMS** ECG, arrhythmia classification, CNN, transformer, Grad-CAM, eXplainable AI, deep learning.

## I. INTRODUCTION

Cardiovascular diseases (CVDs) remain the leading cause of mortality worldwide, accounting for approximately 31% of all global fatalities annually [1]. Among these, arrhythmia is a condition characterized by irregular heart rhythms disrupting the normal electrical conduction of the heart and it is a significant contributor to sudden cardiac death [2]. These

abnormal rhythms range from benign to life-threatening forms such as ventricular tachycardia and fibrillation, which, if undetected and untreated, can lead to cardiac arrest [2], [3]. Electrocardiography (ECG) is the standard non-invasive tool used for detecting arrhythmias by capturing and visualizing the electrical activity of the heart [3], [4]. However, in high-volume clinical environments, manual interpretation of ECG signals is often labor-intensive, prone to diagnostic errors, and not scalable [2]. Furthermore, continuous ECG monitoring produces voluminous and complex signal data

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frequently affected by noise, posing additional challenges for reliable arrhythmia detection [2], [5]. Cardiac arrhythmias denote irregularities in the heart's rhythm and are a significant source of disease and death globally. Among the many arrhythmias, Premature Ventricular Contractions (PVCs) are among the most prevalent ventricular irregularities. PVCs may manifest in healthy persons; nevertheless, recurrent or consecutive PVCs are significantly linked to a heightened risk of cardiomyopathy and abrupt cardiac mortality. The prompt and precise identification of PVCs is therefore of significant therapeutic relevance.

In the field of automated arrhythmia prediction, researchers have often conceptualized the issue as a binary classification challenge, differentiating between normal and arrhythmic beats. This general classification is useful for screening but frequently lacks the specificity required for accurate clinical decision-making, as various arrhythmia types (e.g., PVC, atrial fibrillation, supraventricular tachycardia) exhibit unique morphological patterns and prognostic significance. Arrhythmias are irregularities in cardiac rhythm that vary from benign to potentially fatal, making early and precise identification essential [6]. The electrocardiogram (ECG) is the conventional, non-invasive instrument for documenting cardiac activity; nevertheless, manual interpretation of ECGs is laborious, requires specialized knowledge, and is susceptible to inter-observer variability [7]. To address these constraints, researchers have used Machine Learning (ML) and Deep Learning (DL) for the automation of arrhythmia categorization. Deep learning models, including Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and more recently, Transformers, have shown exceptional performance by autonomously learning intricate ECG features. For instance, deep learning algorithms may acquire both short-term waveform patterns and long-range temporal correlations, facilitating the identification of single-beat irregularities such as Premature Ventricular Contractions (PVCs) and prolonged arrhythmias like atrial fibrillation [7].

Traditional ECG analysis approaches frequently use hand-made time-domain or frequency-domain characteristics, which may not work well with diverse types of patients or signals that are noisy [6]. Additionally, current DL methods for raw 1D signals often struggle with unbalanced data or in understanding their forecasts. One possible option is to turn 1D ECG segments into 2D images, such as waveform plots or time-frequency maps, so that strong image-based DL models may be used. Previous research has demonstrated that CNNs trained on ECG images can do just as well as models that use signals [8]. Zhao et al. have done similar work in which they turned ECG waveforms into 2D images and used an EfficientNet backbone. They got 99.54% accuracy on MIT-BIH data [9]. This backs up the assumption that image representations may accurately show the spatiotemporal structure of an ECG. This study explicitly aims to recognize PVC vs non-PVC beats as a binary

classification issue to overcome this constraint. In contrast to conventional arrhythmia prediction methods, our framework is specifically designed to identify the distinct morphological characteristics of PVCs, hence enhancing diagnostic significance. This study enhances classification accuracy by transforming ECG signals into image representations and using a hybrid EfficientNet–Transformer architecture with Grad-CAM-based interpretability, therefore offering visual explanations that foster confidence and facilitate adoption in clinical practice.

Cardiac arrhythmias remain a significant factor in global cardiovascular morbidity and death. Premature Ventricular Contractions (PVCs) are notably significant owing to their correlation with abrupt cardiac mortality, ventricular dysfunction, and the advancement to more severe arrhythmias. Although many machine learning and deep learning methodologies have been introduced for general arrhythmia identification (i.e., categorizing beats as normal vs arrhythmic), these systems often categorize all irregular rhythms as a single group. This general categorization approach may hide clinically significant differences, particularly when accurate diagnosis of specific arrhythmia subtypes like PVCs is necessary for prompt management.

Furthermore, existing methodologies encounter obstacles, including:

- Skewed datasets, wherein Premature Ventricular Contractions (PVCs) occur significantly less frequently than regular beats, resulting in biased models [10].
- Restricted interpretability, as numerous high-performing deep learning models operate as “non-transparent models,” diminishing clinicians’ trust [11].
- Inadequate feature extraction, as Convolutional Neural Network (CNN)-based models may falter in capturing both localized morphological alterations and extensive temporal dependencies in electrocardiogram (ECG) signals [12].

Consequently, there exists an urgent need for a precise, fast, and interpretable deep learning system that explicitly focuses on the categorization of PVC vs non-PVC, addressing both performance and clinical applicability. Inspired by these advancements, our proposed work transforms single-lead ECG beat data into  $224 \times 224$ -pixel waveform images and employs an innovative hybrid deep learning model for arrhythmia classification. We present ECG-CNNTX (ECG Classification using CNN and Transformer with eXplainability), which integrates an EfficientNetB0 backbone with a Vision-Transformer attention module. This design utilizes CNN layers to extract local waveform information and a Transformer to describe long-range relationships in the ECG image. We utilize gradient-weighted class activation maps (Grad-CAM) to highlight the regions of the ECG image that influence each prediction, thereby enhancing the interpretability of the model.

The primary contributions of this study are as follows:

- 1) Noise-filtered ECG-to-image preprocessing: To capitalize on image-based deep learning techniques, we transform

annotated ECG signals into 2D waveform representations that are normalized and denoised.

2) Hybrid CNN–Transformer model: We develop a fusion model that integrates a Vision Transformer block (for global context) and EfficientNetB0 (for local feature extraction), thereby facilitating the robust detection of arrhythmias.

3) Empirical performance: Our image-based deep learning approach rivals the most advanced signal-based models, as evidenced by achieving better results for PVC detection.

4) Grad-CAM's Explainability: Grad-CAM is employed to emphasize ECG waveform regions (e.g., QRS complexes, ST segments) that influence the model's decision, thereby connecting predictions to medical intuition.

The remainder of this research work is structured as follows. Section II reviews recent research on the classification of ECG arrhythmias. The proposed CNN-Transformer model and the dataset processing (ECG-to-image conversion) are both detailed in Section III. Experimental results and comparisons are presented in Section IV. Section V concludes and delineates potential future research directions.

## II. RELATED WORKS

Arrhythmia detection using Electrocardiogram (ECG) signals has received significant attention recently, particularly with the rise of Deep Learning (DL) techniques. The MIT-BIH Arrhythmia Database has been widely adopted as a standard benchmark for evaluating model performance in this domain [10]. To enhance the robustness and generalization of DL models, preprocessing techniques such as signal normalization and data augmentation have been extensively explored.

Recent advancements in ECG classification have used deep learning to overcome the shortcomings of conventional signal-based techniques. Convolutional neural networks (CNNs) autonomously extract local morphological information; however, their static receptive fields may overlook long-range cardiac patterns [7]. As a result, Transformer topologies, which use self-attention to capture both local and global dependencies, have started to prevail in state-of-the-art methodologies [13], [14]. DeepECG-Net [13] combines a CNN backbone with Transformer layers, attaining better accuracy in general heartbeat abnormality detection. This hybrid approach incorporates denoising and an adaptive learning component, demonstrating robust performance across several arrhythmia datasets [13]. Kim et al. [14] proposed a hybrid CNN-Transformer that first employs the Stockwell time-frequency transform, thereafter using CNNs for local feature extraction and a Transformer for long-term dependency. Their approach achieved 97.8% accuracy on a large arrhythmia dataset and 99.58% on MIT-BIH for five arrhythmia categories, all without necessitating specific R-peak detection [14]. These studies show that the integration of CNNs with transformers enhances performance; nonetheless, they often concentrate on general arrhythmia detection and neglect interpretability.

Transformers have furthermore undergone pre-training on extensive ECG datasets to enhance performance. The HeartBEiT model [15] employs masked image modeling to train a Vision Transformer on 8.5 million ECG waveforms. HeartBEiT, optimized for particular tasks such as cardiomyopathy and infarction, surpassed CNNs in low data scenarios and provided enhanced explainability via the generation of attention maps that emphasize pertinent ECG regions [15]. Nonetheless, HeartBEiT prioritizes clinical diagnosis beyond arrhythmia (e.g., ST-elevation myocardial infarction), and its extensive pretraining renders it less suitable for simpler tasks. Recently, CardioPatternFormer reconceptualized the ECGs as patterns similar to language. It employed a transformer with a pattern-guided attention mechanism to identify diagnostic waveform segments [16]. This approach achieved strong classification performance in intricate multi-pathology ECGs and offers visible attention heatmaps for clinicians. Although CardioPatternFormer enhances interpretability, it functions as a broad ECG classifier, and its performance metrics for particular arrhythmias, such as PVCs, remain unreported.

A prominent theme in contemporary literature is the development of specialized PVC detectors. uPVC-Net is a comprehensive PVC detection model developed using 8.3 million single-lead beats from various sources, including Holter monitors and wearable patches. It employs a bespoke CNN-based architecture with multi-source training, reserving each dataset for assessment. uPVC-Net attained AUCs ranging from 97.8% to 99.1% on previously unencountered datasets, including 99.1% on wearable ECGs [17]. This shows superior generalization across lead configurations and demographic variations. Nonetheless, uPVC-Net is only focused on binary categorization of PVC vs non-PVC and is not intended for multi-class arrhythmia tasks, nor does it prioritize interpretability.

Chen et al. combined Multi-scale Time-Frequency (MTF) encoding with BiFormer, a Transformer architecture that employs bi-level routing attention. It translates single-lead ECGs into spectrogram-like inputs and utilizes sparse attention to extract spatiotemporal ECG characteristics. In the MIT-BIH PVC detection challenge, the MTF+BiFormer model achieved a better accuracy [18], surpassing previous methodologies. The BiFormer's hierarchical attention successfully balances local features and global rhythm patterns. Although this method demonstrates high accuracy, it employs a convoluted multi-stage pipeline and does not provide a precise interpretability analysis beyond acknowledging its attention mechanism [18]. Ebrahimpoor et al. (2024) [19] utilized multi-domain feature extraction (time, frequency, chaos) in conjunction with an autoencoder for dimensionality reduction and SVM, KNN, and NN classifiers. It indicates superior performance compared to previous studies on MIT-BIH; nonetheless, this handcrafted-feature methodology significantly diverges from end-to-end deep learning and fails to provide model explanations or use image representations. Conversely, lightweight variants designed for wearable

implementation have been developed. Busia et al. [20] developed a Tiny Transformer (6k parameters) for the identification of 5-class arrhythmias using the MIT-BIH dataset. This model operates in around 4.28 ms on an ultra-low-power CPU by aggressive pruning and integer quantization, attaining an accuracy of 98.97%. The authors used ECG augmentations to enhance noise resilience, achieving an accuracy of 98.36% after deployment. This study demonstrates that transformer models may be reduced to a size compatible with microcontrollers without significantly compromising accuracy. The disadvantage is that such severe simplicity often excludes any substantive interpretability analysis (no Grad-CAM or attention maps were provided), concentrating only on efficiency. Numerous studies explicitly embrace the ECG-as-image paradigm, akin to our proposed approach. Zhang et al. [21] transform ECG beats into 2D images with a Relative Position Matrix (RPM) encoding and train a residual network with focus on these images. Their RPM-based approach showed superior accuracy in arrhythmia classification, surpassing simpler image transformations. This verifies that meticulously crafted image encodings of ECG may achieve elevated accuracy. Nevertheless, these specialist transformations (e.g., RPM or Stockwell) lack widespread standardization and may not be applicable in general contexts; also, they lack the explainability of their model. Previous studies have demonstrated the efficacy of using generic spectrograms or scalograms as two-dimensional inputs. However, these methods often rely on substantial convolutional neural networks (e.g., EfficientNet, ResNet) that lack a transformer element. Conversely, our methodology employs raw waveform graphs, similar to time-series plots, as images and integrates EfficientNetB0 with a Transformer, thus connecting both concepts. In contrast to previous image-based techniques, we deliberately use Grad-CAM to emphasize ECG areas critical for PVC prediction, therefore effectively improving clinical interpretability. A prevalent shortcoming in the literature is the absence of explicit decision-making logic. Few current ECG models include visualization of learnt characteristics; except for attention maps in CardioPatternFormer and HeartBEiT, most regard the model as a non-transparent model. Furthermore, several cutting-edge techniques are tailored for general arrhythmia or cardiovascular disease activities and may not explicitly focus on premature ventricular contraction detection. Models such as uPVC-Net and BiFormer tackle PVCs; nevertheless, none use a 2D image representation or provide Grad-CAM-like saliency. Furthermore, previous studies often need intricate preparation techniques (such as wavelet transformations or patch embedding) and fail to prioritize resilience against noisy wearable data. The proposed ECG CNNTX model mitigates these deficiencies by using a basic ECG snippet as a visual representation, allowing both CNN and Transformer to use temporal and morphological features. Our technique attains 99% accuracy in PVC identification. It employs Grad-CAM for elucidation, merging the superior efficacy

of hybrid deep models with transparent visual justification, so addressing a critical need in interpretable, image-based ECG analysis.

In conclusion, recent studies have achieved very high arrhythmia detection rates, sometimes exceeding 95%, using CNNs, RNNs, or hybrid architectures. Numerous individuals use either unprocessed signal patterns or programmed transformations as input into CNNs. However, research has not fully leveraged direct ECG waveform images with contemporary vision algorithms. Furthermore, there is a lack of comprehensive visual explainability. To rectify these deficiencies, we propose a methodology that transforms ECG beats into two-dimensional waveform representations and utilizes a CNN-Transformer amalgamation with Grad-CAM. This uses the benefits of image-based CNNs (robust feature extraction) and self-attention while offering visual model interpretation. Our objective is to enhance the efficacy of classification and clinical reliability compared to existing methodologies.

### III. METHODOLOGY

This section provides a detailed description of the methods used in creating an innovative, interpretable deep learning system for arrhythmia identification using ECG data. The proposed approach converts ECG signal segments into images and employs a hybrid model that integrates EfficientNetB0 with a lightweight Transformer attention mechanism, followed by hyperparameter tuning and assessment.

The proposed approach comprises different phases: data gathering, signal segmentation and preprocessing, ECG signal-to-image conversion, data augmentation, model training, and assessment, as shown in Figure 2.

#### A. DATASET DESCRIPTION

ECG images were collected from the MIT-BIH Arrhythmia Database Directory [22]. Each recording was exported as a color image (RGB, arbitrary resolution) that visually encodes the waveform, as shown in Figure 1. Subject demographics, signal features, and the two beat classes used for binary classification (Normal and PVC) are summarized in Table 1 of the MIT-BIH Arrhythmia Database, which was utilized in this study. Clinical specialists supply all beat annotations and include a range of arrhythmic and normal beat classifications. In this study, binary classification is employed to identify Premature Ventricular Contractions (PVCs) [23], labeled as class 1, while all other beats are assigned to class 0.

#### B. ECG SIGNAL SEGMENTATION AND LABELING

Every ECG recording is analyzed with the wfdb library. Beat locations are derived from the respective annotation files (.atr) for each track. A window of 250 samples ( $\pm 125$  samples) is extracted around each indicated beat position to isolate the beat segment. This time equates to around 0.7 seconds of signal, enough to include the whole morphology of the P-QRS-T complex. Each segment is

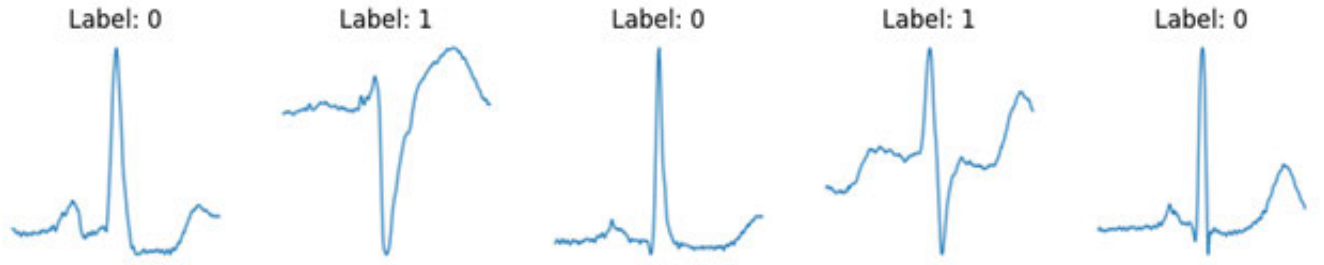


FIGURE 1. Sample ECG images.

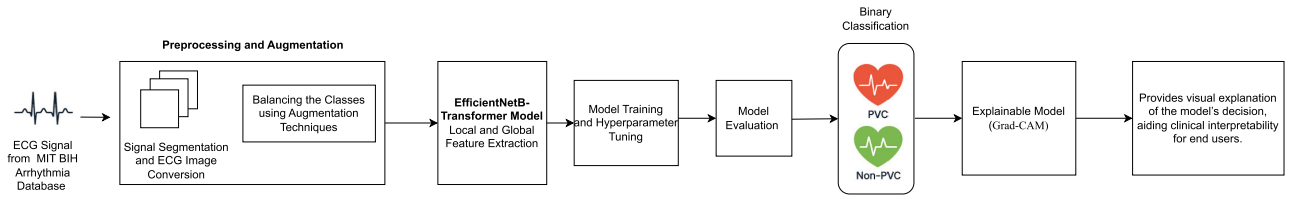


FIGURE 2. Overview of the proposed model.

TABLE 1. MIT-BIH arrhythmia database description.

| Attribute                  | Details                                                                   |
|----------------------------|---------------------------------------------------------------------------|
| Dataset Name               | MIT-BIH Arrhythmia Database                                               |
| Source                     | PhysioNet                                                                 |
| Total Subjects             | 47 (25 male, 22 female)                                                   |
| Age Range                  | 23 – 89 years                                                             |
| Recording Duration         | ~30 minutes per record                                                    |
| Sampling Frequency         | 360 Hz                                                                    |
| ECG Leads Available        | MLII and V1                                                               |
| Lead Used in Study         | MLII                                                                      |
| Number of Records          | 48                                                                        |
| Annotation Source          | Manually annotated by clinical experts                                    |
| Total Beat Types Annotated | Normal, Ventricular Premature (PVC), Left/Right Bundle Branch Block, etc. |
| Focus Beat Classes         | Class 0: Normal Beats; Class 1: Ventricular Ectopic Beats (VEB, 'V')      |

designated as '1' for PVC and '0' for non-PVC, according to the annotation symbol. The segments are cataloged in a list with their corresponding binary labels.

Let  $X_r = \{x_t^r\}_{t=1}^{T_r}$  be the ECG signal from record  $r$ , with annotations  $A_r = \{a_i^r\}_{i=1}^{N_r}$ , where  $a_i^r \in \mathbb{Z}_+$  denotes the sample index of the  $i^{\text{th}}$  annotated beat.

For each beat:

$$s_i^r = X_r[a_i^r - \Delta : a_i^r + \Delta] \in \mathbb{R}^{2\Delta}, \quad (1)$$

where  $\Delta = 125$ , i.e., a 250-sample window centered at the R-peak.

Each segment  $s_i^r$  is labeled as:

$$y_i^r = \begin{cases} 1, & \text{if } l(a_i^r) \in \text{PVC symbols,} \\ 0, & \text{otherwise.} \end{cases} \quad (2)$$

The complete labeled dataset is then constructed as:

$$D = \{(s_i^r, y_i^r) \mid r = 1, \dots, R; i = 1, \dots, N_r\}. \quad (3)$$

For notational simplicity, all segments across records are enumerated by a single index  $j$ , yielding:

$$D = \{(s_j, y_j)\}_{j=1}^N, \quad (4)$$

where  $N = \sum_{r=1}^R N_r$  and each pair  $(s_j, y_j)$  corresponds to one  $(s_i^r, y_i^r)$ .

#### Algorithm 1 Conversion of 1D ECG Signals to 2D Image Representations

- Data Loading:** Import ECG signals and annotations from the MIT-BIH Arrhythmia Database using WFDB.
- Segmentation:** Extract 250-sample windows ( $\pm 125$  around R-peak) from Lead II to capture complete P-QRS-T morphology.
- Filtering:** Apply a 4th-order Butterworth bandpass filter (0.5–40 Hz) to remove baseline wander and high-frequency noise.
- Normalization:** Perform median-centered amplitude normalization to ensure consistent waveform scaling.
- Plotting:** Visualize each normalized segment as a clean waveform (no axes or grids) using Matplotlib.
- Image Processing:** Save plots as PNGs, convert to RGB, and resize to  $224 \times 224$  pixels using bicubic interpolation.
- Dataset Preparation:** Organize images into class folders (Normal, Ventricular, etc.) and rescale pixel values (1/255) for model input.

The wfdb Python library is employed to read raw ECG recordings. Beats are segmented into fixed-length windows of 250 samples (approximately 0.7 seconds) centered around each annotated R-peak, and annotations are extracted for each recording. The segmentation guarantees that each sample accurately represents the cardiac cycle's morphology, which encompasses P waves, QRS complexes, and T waves. Beat markings are allocated according to whether the annotation symbol corresponds to a PVC. This procedure transforms each unprocessed recording into a collection of 1D signal segments that are labeled and suitable for subsequent processing.

### C. ECG SIGNAL TO IMAGE TRANSFORMATION

This study transforms 1D ECG signal segments into 2D image representations, facilitating the use of advanced image-based deep learning architectures, as shown in Figure 2. Each 250-sample signal segment is shown using matplotlib, with the waveform rendered on a white backdrop, devoid of axis labels, ticks, and gridlines to minimize distraction. The visual representation is then transformed into a high-resolution RGB image via the Python Imaging Library (PIL). The plots are stored as PNG images and scaled to  $224 \times 224$  pixels, a typical input resolution for most convolutional neural networks, including EfficientNet. The images are then organized into named subdirectories (training/testing, class 0/1), facilitating compatibility with conventional image classification algorithms. This innovative signal-to-image methodology utilizes robust computer vision models that have been pre-trained on natural image datasets, enabling the extraction of spatial features from ECG waveforms. This transformation turns ECG signals into a format suitable for image classification, facilitating the use of pretrained models on medical signal data and establishing an image-centric approach for ECG interpretation, as shown in Algorithm 1. After converting the ECG signal to an image, the final dataset included 112,559 images in total. These were split into two classes: Class 0 (Normal) had 105,430 images and Class 1 (Abnormal / Ventricular) had 7,129 images. To keep the class balance, the dataset was split into 80% for training (90,047 images) and 20% for testing (22,512 images). We used a fixed random seed (random\_state = 42) for all of the trials to make sure that the stated findings could be reproduced completely.

### D. PRE-PROCESSING AND AUGMENTATION

Preceding image conversion, each ECG segment underwent amplitude normalization and noise filtration to enhance signal quality and stability. Baseline wander and high-frequency noise were suppressed using standard bandpass filtering (0.5–40 Hz), ensuring that the P–QRS–T morphology remained intact. This filtering step minimizes motion artifacts and powerline interference, thereby improving the clarity of waveform images and reducing model sensitivity to noise variations. All samples are resized to  $224 \times 224$  pixels and rescaled to [0,1] by the generator (rescale = 1/255).

Random geometric augmentation was intentionally disabled to preserve the strict spatial relationships between P, QRS, and T complexes; earlier pilot experiments showed that aggressive affine transforms distorted clinically relevant morphology. Nevertheless, minor color jitter ( $\pm 5\%$  channel-wise Gaussian noise) is implicitly injected by mixed precision training, providing mild regularization without corrupting the underlying waveform.

The ECG images are subjected to image augmentations to enhance generalization and diversify the training samples. The training data is augmented by using TensorFlow's ImageDataGenerator. This augmentation replicates the real-world variability in ECG morphology caused by lead misplacements, physiological differences, or signal distortions, without compromising the clinical information incorporated in the waveform. This augmentation method, although visually inconspicuous, improves the model's generalization capacity without compromising clinically significant waveform data. This approach diverges from conventional ECG augmentation methods (such as temporal warping or signal noise injection) and uses transformer-based algorithms tailored for biomedical waveform shape.

### E. PROPOSED MODEL ARCHITECTURE

The proposed technique entails preprocessing ECG signals from the MIT-BIH dataset by transforming them into images and implementing augmentation for class balance, then using a hybrid EfficientNetB0–Transformer model for binary classification. Grad-CAM elucidates model choices for clinical interpretability. Early detection of arrhythmia from Electrocardiogram (ECG) imagery remains challenging due to (i) subtle visual differences among classes, (ii) scarcity of labelled data at scale, and (iii) the need for low-latency inference on edge devices. Convolutional Neural Networks (CNNs) provide strong inductive biases for local pattern recognition, yet may fail to capture long-range dependencies that differentiate morphologically similar rhythms.

The classification model's foundation is a streamlined hybrid architecture termed ECG-CNNTX, which integrates EfficientNetB0 for local feature extraction and a Transformer block for modeling global contextual interactions. The concept is fundamentally based on a hybrid design that combines EfficientNetB0 with a lightweight Transformer block. EfficientNetB0 is chosen as the foundational convolutional network because of its ideal equilibrium between model size and accuracy. It is initialized with pre-trained ImageNet weights, and its layers are frozen to maintain the learned representations. The output from the global pooling layer of EfficientNetB0 is restructured into a single-token sequence and input into a custom-designed transformer block [24].

EfficientNetB0 is a convolutional neural network architecture recognized for its efficiency and superior performance. It is used as a frozen feature extractor, pre-trained on ImageNet and devoid of its classification head. The result is globally aggregated to generate a 1280-dimensional

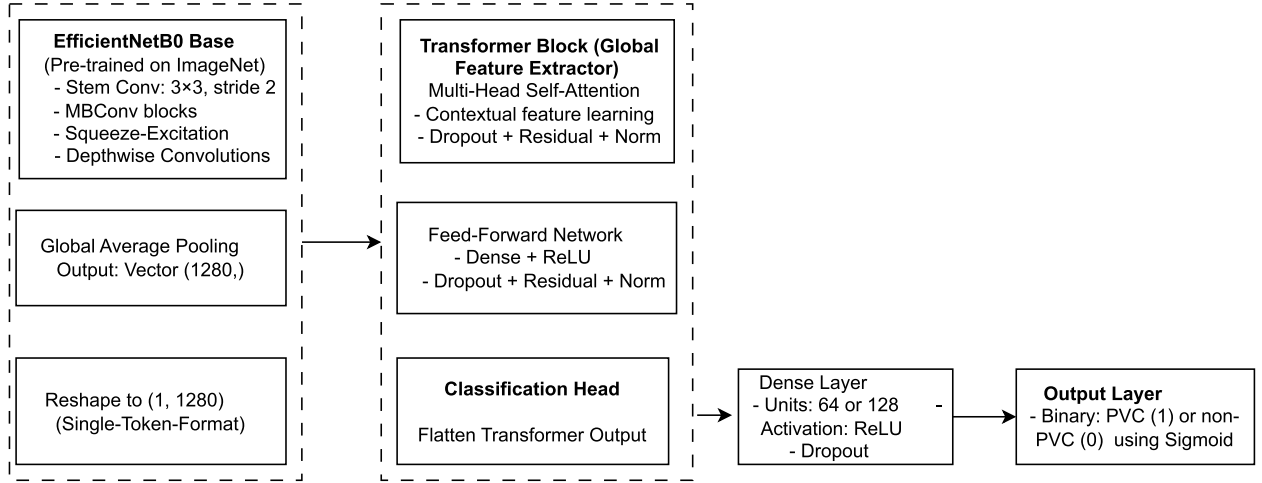


FIGURE 3. ECG-CNNIX hybrid architecture.

feature vector. The vector is restructured and input into a single-layer transformer encoder block that includes a multi-head self-attention mechanism, including two attention heads. This integration enables the model to capture both intricate morphological details and extensive structural links among features, surpassing the limitations of models that rely solely on CNNs or Transformers, as shown in Figure 3. Transformers, conversely, excel at global context modeling but are parameter-heavy and data-hungry [25]. We therefore propose a hybrid architecture that integrates an EfficientNetB0 backbone with a single-layer, lightweight Transformer self-attention head. This study uses ImageNet priors, injects global context at minimal cost, and preserves a lean computational footprint ( 5M trainable parameters), thereby addressing the limitations in existing research works. Transformers have demonstrated state-of-the-art results in temporal and medical imaging tasks [26]. To limit the parameter complexity, we deploy a single-layer Multi-Head Self-Attention (MHSA) with two heads (key\_dim = 1280). The Transformer block comprises a multi-head self-attention layer, succeeded by layer normalization, dropout, and a feed-forward dense network. The attention mechanism enables the model to comprehend inter-feature interactions inside the visual embedding, emulating the temporal attention seen in real sequences [21]. The application of attention to spatial embeddings derived from ECG waveforms provides a robust dynamic modeling capability, improving the model's capacity to identify minor morphological alterations suggestive of PVCs. The CNN captures fine-grained morphology (P-waves, QRS complexes), while the transformer head models the relative timing and amplitude relations across the entire frame [27]. This interaction boosts sensitivity to rhythm class nuances that are otherwise masked in purely convolutional representations. The mathematical analysis of the MHSA is as follows,

Given image  $\tilde{I}_j$ :

$$z_j = f_{\text{cnn}}(\tilde{I}_j) \in \mathbb{R}^{1280} \quad (5)$$

Reshape:

$$z'_j = z_j[\text{None}, :] \in \mathbb{R}^{1 \times 1280} \quad (6)$$

*Multi-Head Self-Attention (MHSA):*

We define multi-head self-attention for a single-token input. Let  $Q, K, V \in \mathbb{R}^{1 \times d}$  be the query, key, and value vectors. For head  $i$ , attention is computed as:

$$\text{head}_i = \text{Softmax} \left( \frac{QW_i^Q (KW_i^K)^T}{\sqrt{d_k}} \right) VW_i^V \quad (7)$$

The final MHSA output is:

$$\text{MHSA}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h) W^O \quad (8)$$

Then, the transformer-enhanced embedding is:

$$h_j = f_{\text{attn}}(z'_j) = \text{LN}(z'_j + \text{MHSA}(z'_j, z'_j, z'_j)) \in \mathbb{R}^{1 \times 1280} \quad (9)$$

*Classification Head and Binary Output:*

For binary classification, we use the *sigmoid activation function* (not softmax), which maps scalar output to probability:

$$\hat{y}_j = \sigma(W_c h_j + b_c) \in [0, 1] \quad (10)$$

where

$$\sigma(x) = \frac{1}{1 + e^{-x}} \quad (11)$$

is the sigmoid function. This value represents the probability that the beat is a PVC (class 1).

## F. MODEL TRAINING AND EVALUATION

To fine-tune the model architecture and training parameters, we use the Keras Tuner package with Random Search. A small number of trials were sufficient to identify a well-performing configuration that balances complexity with accuracy. This automated tuning process ensures that model parameters are optimized based on features, rather than being manually selected. The final model is trained via

the Adam optimizer with categorical cross-entropy as the loss function. Training is conducted with a batch size of 32 and a learning rate determined via hyperparameter optimization. Early halting is implemented with a patience of 10 epochs, contingent on validation correctness, to avoid overfitting. Dropout is used at both the transformer output and the dense head, augmenting regularization and reducing dependence on particular characteristics. The model is trained using GPU hardware to accelerate convergence and enhance computational efficiency. The binary cross-entropy loss is described as follows:

$$\mathcal{L} = - \sum_{j=1}^N [y_j \log(\hat{y}_j) + (1 - y_j) \log(1 - \hat{y}_j)] \quad (12)$$

The model is optimized using Adam with tuned learning rates (1e-3, 1e-4), dropout **0.3–0.5**, and early stopping on validation accuracy.

#### 1) HYPERPARAMETER OPTIMIZATION

We encapsulate the architecture in a Keras Tuner HyperModel and conduct Random Search over the following space:

- Dense units: 64, 128
- Optimizer: Adam ( $\alpha = 0.9$ ,  $\beta = 0.999$ )
- Learning Rate: Tuned across 1e-3, 1e-4 with cosine annealing in extended runs.
- Early Stopping: Patience = 10 on val\_accuracy to prevent overfitting.
- Dropout: (0.3–0.5) at the dense head and 0.1 within the attention block.

#### G. PERFORMANCE METRICS

The performance assessment utilizes the validation dataset, with accuracy as the primary measure. The efficacy of a classification model is determined by the values of several relevant metrics. Metrics for evaluating performance assist in analyzing the model's efficacy. These facilitate the differentiation of other classification models and the identification of the ideal model for each parameter. Performance metrics such as the confusion matrix, recall, precision, F1 score, and accuracy are often used. The Confusion Matrix is a simple tool that facilitates the evaluation of a model's accuracy. Statistical metrics of accuracy, precision, recall, and F1 score were used to assess the efficacy of classification algorithms applied to the PD voice dataset. The methodology for calculating classified instances is delineated in equations (1) to (4), wherein accurately categorized cases are denoted as TP (True Positive) and TN (True Negative). Conversely, misclassified examples are denoted as FP (False Positive) and FN (False Negative).

$$\text{Accuracy} = \frac{TN + TP}{TN + TP + FN + FP} \quad (13)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (14)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (15)$$

$$F_1 \text{ score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (16)$$

The model attains around 99% classification accuracy in PVC detection, demonstrating its proficiency in differentiating between normal and ectopic beats with considerable precision. Subsequently, the sensitivity, specificity, and F1-score were calculated for a more thorough assessment [28].

Algorithm 2 incorporates beat-wise segmentation, signal-to-image transformation, and local-global attention mechanisms within a unified and explainable framework. This phased algorithmic breakdown offers a logical and research-friendly structure for the proposed model.

#### H. EXPLAINABILITY

We leverage Grad CAM on the penultimate EfficientNet convolutional layer to highlight waveform regions influencing predictions, and overlay attention weight heat maps to expose global token interactions. This dual-view transparency is critical for clinician trust.

Grad-CAM is applied to the final convolutional layer in EfficientNetB0. Given a target class  $c$ , the heatmap  $M^c \in \mathbb{R}^{H \times W}$  is computed as:

$$M^c = \text{ReLU} \left( \sum_k \alpha_k^c A^k \right) \quad (17)$$

where  $A^k$  is the  $k$ -th activation map and

$$\alpha_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k} \quad (18)$$

Interpretability is facilitated by Gradient-weighted Class Activation Mapping (Grad-CAM). This method emphasizes the areas of the input image that significantly impact the model's prediction by back-propagating class-specific gradients via the convolutional layers. When used on ECG images, Grad-CAM identifies the components of the beat (e.g., QRS complex, T-wave) that significantly influence the classification choice. This stage is crucial in medical AI to guarantee transparency, reliability, and congruence with clinical reasoning, hence enhancing acceptance among practitioners [29].

In summary, the proposed technique presents an innovative pipeline for arrhythmia identification that converts 1D ECG data into 2D images, facilitating the use of advanced visual deep learning models. This hybrid architecture, along with hyperparameter optimization and interpretability tools, distinguishes this work from existing literature and offers a robust, explainable, and scalable method for automated ECG analysis. This work presents innovation on several levels, such as the Transformation of signals into images using a scalable, visual format for deep learning. Augmentations implemented at the image level to enhance ECG beat representation diversity. Hybrid CNN-Transformer architecture including spatial encoding and attention-driven temporal dynamics. This pipeline diverges from traditional 1D signal processing, offering a flexible and modular visual framework

**Algorithm 2** Pseudocode of the Proposed Model**Require:** Annotated ECG data from the MIT-BIH Arrhythmia Database**Ensure:** Predicted class for each beat: PVC = 1, non-PVC = 0

- 1: **Phase 1: Conversion of 1D ECG Signals to 2D Image Representations**
  - 1.1 Load ECG signals and annotations from the MIT-BIH Arrhythmia Database using WFDB.
  - 1.2 Segment each ECG record into 250-sample windows ( $\pm 125$  around R-peak) from Lead II.
  - 1.3 Apply a 4<sup>th</sup>-order Butterworth bandpass filter (0.5–40 Hz) to remove noise and baseline wander.
  - 1.4 Normalize each segment by subtracting its median and dividing by peak-to-peak amplitude.
  - 1.5 Plot each normalized waveform on a clean background (no axes or grids) using Matplotlib.
  - 1.6 Save plots as PNGs, convert to RGB, and resize to  $224 \times 224$  pixels via bicubic interpolation.
  - 1.7 Organize images into class folders (Normal, Ventricular, etc.) and rescale pixel values to [0, 1].
  - 3.4 Apply data augmentation (zoom, shift) to training images.
  - 3.5 Normalize pixel intensities to the range [0, 1].
- 2: **Phase 2: Extraction of Features and Learning of Attention**
  - 2.1 Feed input images into EfficientNet-B0 and extract visual embeddings.
  - 2.2 Reshape embeddings into a token sequence suitable for transformer input.
  - 2.3 Process tokens with a multi-head self-attention module.
  - 2.4 Apply layer normalization and a position-wise feed-forward network.
- 3: **Phase 3: Categorization and Output Production**
  - 3.1 Pass the attention-enhanced feature vector through fully-connected (dense) layers.
  - 3.2 Use a sigmoid activation on the final unit to obtain the PVC probability score.
  - 3.3 Threshold the probability (e.g., 0.5) to produce the binary prediction: PVC (1) or non-PVC (0).
- 4: **Phase 4: Assessment and Interpretability**
  - 4.1 Evaluate the model on the test set using classification metrics (accuracy, precision, recall, F1, AUC).
  - 4.2 Apply Grad-CAM on selected images to generate heatmaps highlighting decision-critical regions.
  - 4.3 Visualize and report interpretability results alongside quantitative metrics.
- 5: **return** Predictions and evaluation artifacts (confusion matrix, metrics, Grad-CAM heatmaps)=0

for arrhythmia identification that can be applied to other biological signal areas.

The overall novelty of the proposed work is fourfold. Initially, it presents a novel visual interpretation of ECG data by transforming one-dimensional beats into two-dimensional RGB visuals for vision-based modeling. Secondly, it implements image-space augmentations particularly designed for variations in ECG shape. Third, it introduces a hybrid convolution-attention model that integrates EfficientNetB0 and Transformer blocks for enhanced feature extraction and contextual modeling. Finally, it integrates explainability using Grad-CAM to guarantee transparent decision-making by the model in clinical applications. These contributions provide an innovative and efficient approach for automatic arrhythmia identification.

#### IV. RESULTS AND DISCUSSION

This section delineates the empirical assessment of the proposed ECG-CNNTX model using the MIT-BIH Arrhythmia Database for the binary classification task of differentiating Premature Ventricular Contractions (PVCs) from non-PVC beats. The findings are categorized into four principal components: performance metrics of the model, interpretability analysis using Grad-CAM, ablation study of model components, and comparative benchmarking with existing

methodologies. All experiments were conducted using TensorFlow 2.19 in conjunction with the Keras API for high-level model construction. Computation was performed on Google Colab Pro, utilizing an NVIDIA T4 GPU with 15 GB VRAM, supported by two virtual CPUs and 13 GB of RAM. The training of the proposed ECG-CNNTX hybrid model was executed using conventional deep learning methodologies on a GPU-accelerated environment. Table 2 delineates the comprehensive training configuration, including optimization settings, learning rate scheduling, loss functions, and hardware specifications used to guarantee steady and efficient convergence during the training process.

The ECG-CNNTX model was assessed using a separate test set consisting of ECG images obtained from the MIT-BIH Arrhythmia dataset. The model attained a classification accuracy of 99%, exhibiting strong proficiency in differentiating PVCs from non-PVC beats. The ECG-CNNTX model exhibited consistently superior performance across critical assessment criteria, indicating its robust capacity to distinguish between normal and PVC heartbeats properly. The equilibrium between accuracy and recall signifies dependable and resilient predictions, while the convergence patterns and operational efficiency underscore the model's computational efficacy. Table 3 findings confirm the model's capability for real-time and clinically relevant ECG arrhythmia identification.

TABLE 2. Training setup for ECG-CNNTX model.

| Parameter                        | Configuration                    | Description                                                                                                                  |
|----------------------------------|----------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Dense units (in classifier head) | 128                              | Provided higher representational capacity without overfitting, selected from {64, 128} based on validation accuracy.         |
| Dropout rate (dense head)        | 0.4                              | Balanced regularization and convergence stability, tuned in the range 0.3–0.5 using validation loss.                         |
| Dropout rate (attention block)   | 0.1 (fixed)                      | Lightweight regularization within the MHSA block, retained constant during optimization.                                     |
| Optimizer                        | Adam                             | Adaptive Moment Estimation for faster convergence and dynamic learning rate [1].                                             |
| Attention Heads (MHSA)           | 2 heads                          | Selected from {1, 2} based on validation accuracy; using two heads improved contextual representation of waveform relations. |
| Learning Rate                    | 1e-3 to 1e-4 (tuned)             | LR range tuned empirically to balance convergence speed and stability.                                                       |
| Loss Function                    | Binary Cross-Entropy             | Suitable for binary classification tasks such as PVC vs. non-PVC detection.                                                  |
| Batch Size                       | 32                               | Ensures sufficient gradient stability with manageable memory usage.                                                          |
| Early Stopping                   | Patience = 10                    | Stops training if no improvement in validation loss for 10 consecutive epochs.                                               |
| Epochs                           | 20                               | Maximum number of training iterations (with early stopping applied).                                                         |
| Hardware Platform                | Google Colab Pro (NVIDIA T4 GPU) | Ensures accelerated training using high-performance cloud GPU infrastructure.                                                |

TABLE 3. Performance analysis of the proposed model.

| Metric               | Value (%) |
|----------------------|-----------|
| Accuracy             | 99.00     |
| Precision            | 98.90     |
| Recall (Sensitivity) | 99.00     |
| F1-Score             | 98.95     |
| AUC-ROC              | 99.70     |

A series of hyperparameter tuning experiments was undertaken using KerasTuner’s random search approach to investigate a better configuration of the proposed ECG-CNNTX architecture. The tuning procedure sought to assess the impact of various combinations of attention heads, dense layer sizes, feed-forward dimensions, dropout rates, and learning rates on model convergence, generalization, and computing efficiency. Each trial was conducted for 20 epochs using uniform data and augmentation parameters, hence assuring evaluative consistency. Table 4 presents the performance, training stability, and execution time throughout these trials.

Figure 4 is a confusion matrix that illustrates the robust classification efficacy of the proposed ECG-CNNTX model in distinguishing between Normal and PVC heartbeats. A substantial quantity of true positives and true negatives was noted, with only negligible incorrect classifications, demonstrating that the model effectively distinguishes between both groups with elevated sensitivity and specificity. The low error rate is especially important in clinical environments, where failing to detect a PVC event (false negative) or incorrectly identifying a normal beat (false positive) may result in erroneous diagnosis or therapy. The ROC curve

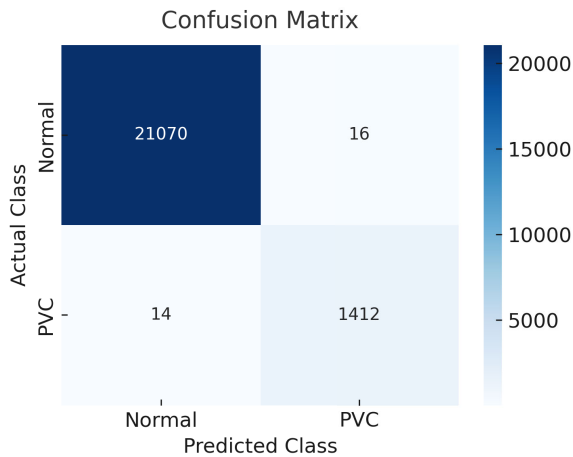


FIGURE 4. Confusion matrix of the proposed model.

further substantiates the model’s resilience, producing an Area Under the Curve (AUC) of roughly 0.997, as shown in Figure 5. The almost flawless AUC indicates that the model has outstanding discriminating capability across many categorization thresholds, hence affirming its dependability despite minor ECG fluctuations. The smooth convexity of the ROC curve indicates the model’s uniform predictive confidence for both classes. The integrated findings from the confusion matrix and ROC analysis validate that the hybrid design proficiently captures both spatial and temporal data, attaining a high degree of accuracy, generalization, and clinical significance. The accuracy and loss curves for training and validation were studied over all epochs to address model convergence and assess possible overfitting. The training and validation accuracies converged close to 99%, and that the

**TABLE 4.** Trial summary with hyperparameters and performance.

| Trial   | Execution Time | Best Validation Accuracy | Training Accuracy | Loss   |
|---------|----------------|--------------------------|-------------------|--------|
| Trial 1 | 19m 02s        | 0.99                     | 0.9450            | 2.0152 |
| Trial 2 | 19m 18s        | 0.99                     | 0.9428            | 2.4009 |
| Trial 3 | 18m 45s        | 0.99                     | 0.9577            | 1.0368 |

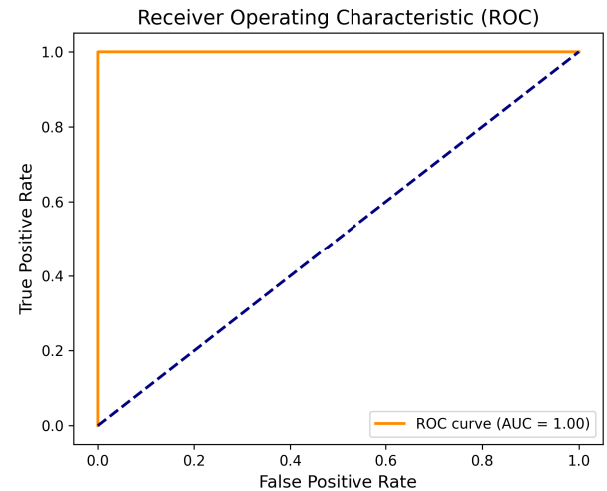
loss curves for the two sets of data fell linearly without any divergence. Both curves are trending in the same direction, which means that the ECG-CNNTX model was able to reach stable optimization and maintain its generalization during training. Results showing no oscillations or increasing gaps between training and validation performance indicate that the regularization tactics used, which included early halting, dropout, and considerable data augmentation, successfully reduced the risk of overfitting. These results confirm the model's resilience in real-world ECG classification settings and its dependable convergence behavior.

The superior performance of the proposed ECG-CNNTX model is mostly attributable to its combined preprocessing and hybrid architectural framework. The use of amplitude normalization and noise filtering prior to ECG-to-image conversion guarantees the preservation of clinically significant P–QRS–T morphology, yielding clean and uniform input for feature extraction. In the hybrid architecture, the EfficientNetB0 backbone adeptly captures intricate local properties, including QRS width and ST-segment deviations, while the Transformer block models extensive contextual relationships across waveform regions. This collaborative representation enables accurate differentiation of nuanced PVC morphologies that single-stream CNN or ViT models cannot identify. Misclassification analysis indicated that residual mistakes mostly arose from borderline or noise-distorted PVC beats characterized by low signal amplitude, suggesting that enhanced adaptive filtering or attention refinement might enhance detection consistency. The integration of signal-level denoising and architectural synergy improves the model's resilience and interpretability, showcasing distinct benefits compared to baseline CNN and ViT setups.

#### A. ABLATION STUDY

Ablation research was conducted to examine the individual contributions of the EfficientNetB0 and Transformer components. The findings are contained in Table 5. The integration of EfficientNetB0 for local feature extraction with the Transformer for global context modeling significantly surpasses the performance of individual baselines, as shown in Table 5. The hybrid design leverages spatial fidelity using convolutional neural networks and models temporal dependencies using attention techniques.

The reason for the relatively reduced accuracy of the standalone EfficientNetB0 and Transformer models is that each model independently captures only a specific representation. EfficientNetB0 concentrates on local morphological patterns,

**FIGURE 5.** ROC analysis of the proposed model.

while the Transformer models long-range dependencies. The proposed hybrid ECG-CNNTX model outperforms its competitors because it makes use of these complementing feature domains, which boost discriminative capabilities. Furthermore, in contrast to previous research that found multi-class ECG issues, the inter-class changes in the current study are morphologically minor since it deals with a binary PVC vs. non-PVC job. Regardless, the state-of-the-art comparison in Table 6 further validates that the proposed model provides better generalization and competitive accuracy. To improve class balance and robustness, we applied conservative geometric augmentations to training images: zoom ( $\pm 10\%$ ), width shift ( $\pm 10\%$ ), and height shift ( $\pm 10\%$ ), as shown in table 6. Horizontal flips were considered but only applied after validating they did not invert diagnostic polarity in our plotted waveforms; rotations and vertical flips were deliberately excluded to avoid inverting time or polarity. An ablation comparing models trained with and without the above geometric augmentations shows that these conservative transforms improved test generalization (reduced overfitting) while preserving waveform morphology. The ablation findings demonstrate that ImageNet pretraining substantially improves model performance and accelerates convergence speed. The pretrained (PT) variation attained an accuracy of 99%, surpassing the 95.12% reached with Random Initialization (RI), coupled with steady improvements in precision, recall, and F1-score. This illustrates that transfer

**TABLE 5. Ablation study – model component contribution.**

| Architecture Variant                     | Accuracy | Precision | Recall | F1-score |
|------------------------------------------|----------|-----------|--------|----------|
| EfficientNetB0 only                      | 95.23%   | 94.89%    | 94.12% | 94.50%   |
| Transformer only (ViT base model)        | 94.12%   | 93.87%    | 92.56% | 93.21%   |
| ECG-CNNTX (EfficientNetB0 + Transformer) | 99.01%   | 98.93%    | 98.85% | 98.89%   |

**TABLE 6. Ablation study on the effect of geometric augmentation.**

| Augmentation Applied                                   | Accuracy (%) | Precision | Recall | F1-Score | Observation                                                                                                     |
|--------------------------------------------------------|--------------|-----------|--------|----------|-----------------------------------------------------------------------------------------------------------------|
| None (Baseline)                                        | 97.6         | 0.972     | 0.971  | 0.971    | Slightly lower generalization and mild overfitting observed due to limited data variability.                    |
| Geometric Augmentation (Proposed Model) (zoom + shift) | 99.0         | 0.989     | 0.990  | 0.989    | Matches the final model performance (Table 3); improved robustness and preservation of ECG waveform morphology. |

**TABLE 7. Ablation study on the impact of imagenet pretraining.**

| Variant | Pretraining | Acc. (%)     | Prec. (%) | Rec. (%) | F1 (%)  | AUC   |
|---------|-------------|--------------|-----------|----------|---------|-------|
| (a) RI  | None        | 95.12 ± 0.45 | 94.76     | 94.81    | 94.78   | 0.962 |
| (b) PT  | ImageNet    | 99.00 ± 0.12 | 98.90     | 99.02    | 98.9599 | 0.997 |

learning allows the model to use generic visual cues, enhancing generalization and decreasing training epochs, as seen in table 7.

## B. COMPREHENSIVE COMPARATIVE ANALYSIS USING BENCHMARK DATASET

Table 8 covers recent deep learning models for ECG classification, namely targeting PVC detection and general arrhythmias. For each model, we specify the input type, reported accuracy (usually on MIT-BIH or similar datasets), whether the ECG signal was transformed into an image, the application of data augmentation, the classification as a hybrid CNN+Transformer model, the distinction between PVC and non-PVC binary classifiers, and the use of explainability techniques.

Recent models either function on unprocessed ECG data or transform them into 2D images (such as spectrograms, scalograms, or other encodings) to use CNN/Transformer architectures. For instance, ECGConVT and uPVC-Net convert data into visual representations, whereas CardioPatternFormer and Tiny Transformer handle 1D signals directly. The most elevated documented accuracies in PVC identification are attributed to signal-to-image methodologies: BiFormer (utilizing MTF images) attained around 99.45% accuracy in distinguishing PVC from non-PVC [18], while uPVC-Net (employing STFT) obtained around 97.8% [17]. General arrhythmia models such as ECGConVT and Tiny Transformer achieve better accuracy on the MIT-BIH dataset [30]. ECGConVT is the only hybrid model that integrates a CNN backbone with a Vision Transformer module; the other models are exclusively transformers or RNNs. Data augmentation was explicitly used in the Tiny Transformer model

(noise/artifact augmentation) [20], but not documented in the others. Explainability, such as attention or saliency maps, is typically uncommon; CardioPatternFormer and HeartBEiT integrate attention-based visualizations, whilst the others do not disclose particular XAI methodologies. Although some models provide slightly superior accuracy, the majority are deficient in interpretability and rely on signal-based inputs. Conversely, the suggested ECG-CNNTX utilizes an image-based methodology and incorporates explainability, making it more appropriate for clinician-oriented applications. The attention heatmap primarily focuses on the QRS complexes, confirming that the model acquires clinically relevant features.

Explainability is a vital component of AI-based ECG analysis, especially when using deep architectures like Transformers. This bolsters the overarching initiative for transparent and responsible AI systems in healthcare [31]. Grad-CAM visualizations were used to improve clinical interpretability. Grad-CAM was used on ECG waveform images to pinpoint locations that significantly influence model decision-making. The attention heatmaps consistently highlighted the QRS complex, correlating with established morphological indicators of PVCs. Warmer regions correlate with places that the model prioritizes when detecting PVCs. The model primarily focuses on the R-peaks and atypical depolarization regions suggestive of ectopic ventricular activity.

## C. DISCUSSION

The ECG-CNNTX model represents a notable progression in automated arrhythmia identification by integrating a hybrid architecture that combines EfficientNetB0 with a Transformer attention module. This model's primary benefit is its exceptional classification accuracy, attaining 99% in the identification of PVCs, bolstered by robust F1-score and AUC metrics. This result validates the model's efficacy in identifying the subtle waveform changes that differentiate premature ventricular contractions from regular

**TABLE 8.** Comparison of existing models with proposed model.

| Model                     | Input Type                                 | Accuracy (%)                    | Signal-to-Image | Data Augmentation        | Hybrid (CNN+Transformer)   | PVC Binary | Explainability |
|---------------------------|--------------------------------------------|---------------------------------|-----------------|--------------------------|----------------------------|------------|----------------|
| ECGConVT[30]              | 12-lead ECG waveform images                | 98.5                            | Yes             | Not reported             | Yes (CNN+ViT)              | No         | None reported  |
| uPVC-Net[17]              | STFT mel-spectrogram of 1D ECG signal      | 97.83                           | Yes             | Not reported             | No (Bi-GRU RNN)            | Yes        | None reported  |
| BiFormer[18]              | MTF image (Markov Transition Field) of ECG | 99.45                           | Yes             | Not reported             | No (Transformer)           | Yes        | None reported  |
| Tiny Transformer[20]      | Raw ECG signal (1D)                        | 98.97                           | No              | Yes (noise augmentation) | No (Transformer)           | No         | None reported  |
| HeartBEiT[15]             | ECG waveform image (vision transformer)    | Not reported (foundation model) | Yes             | Not reported             | No (Transformer)           | No         | Attention maps |
| ECG-CNNTX(Proposed Model) | ECG images                                 | 99.01                           | Yes             | Yes                      | EfficientNetB0+Transformer | Yes        | Yes            |

beats. Elevated accuracy and recall measures are especially critical in healthcare contexts where misclassification may result in erroneous or postponed diagnoses. The findings indicate the model's capacity to identify minor waveform differences that signify arrhythmic occurrences, even under conditions of data imbalance. The use of EfficientNetB0 and a lightweight Transformer block enhances the performance of the proposed work. This hybrid approach allows the model to concurrently learn local morphological features (e.g., QRS complex, ST segment) and record global temporal linkages throughout the waveform, which are essential for precise ECG interpretation. In contrast to only signal-based approaches, converting ECG data into 2D images enables the system to use robust pretrained vision models, hence diminishing the need for manually constructed features or intricate domain-specific filters. The proposed ECG-CNNTX model exhibits a promising hybrid architecture that effectively combines the feature extraction capabilities of EfficientNetB0 with the global contextual modeling of a lightweight Transformer block. The integration of convolutional and attention-based methods enables the model to accurately distinguish between PVC and normal ECG waveforms, which are then precisely converted into 2D image formats. The proposed model's primary strength is its exceptional generalization performance, shown by a continuously high validation accuracy of 99% across all hyperparameter trials. Trial 3 proved to be the most efficient design, earning the lowest training loss (1.0368) and a competitive training accuracy of 95.77%, in addition to the quickest execution time. This outcome underscores that the model not only acquires knowledge proficiently but also does so with computational efficiency, an imperative criterion for real-time clinical applications.

Ultimately, the data preparation pipeline is designed for scalability and clinical applicability. The transformation of raw ECG segments into images is executed using standard libraries (e.g., WFDB, matplotlib, and PIL) and augmentation techniques such as shifting and zooming. These augmentations enhance robustness without adding artifacts that may confuse the model or mislead physicians. Finally, despite the execution time per trial being less than 20 minutes,

real-time deployment in edge or embedded systems needs further pruning, quantization, or optimization utilizing frameworks such as TensorFlow Lite or ONNX.

The model's training and assessment have only used the MIT-BIH Arrhythmia Database, a commonly employed albeit somewhat constrained benchmark dataset. To guarantee robustness and generalizability, it is essential to assess the model using external datasets, such as PTB-XL, INCART, or real-world hospital-acquired ECGs, which may differ in demographic diversity, acquisition noise, sampling frequency, and annotation processes. In the absence of such validation, the model's efficacy across various clinical populations and hardware configurations cannot be assured. Eventually, the approach presupposes reasonably untainted incoming data. Despite the inclusion of normalization and visual augmentation in the preprocessing phases, the system has not undergone thorough testing in environments characterized by significant signal noise, motion artifacts, or absent beat segments—issues often encountered in ambulatory ECG monitoring. Subsequent versions must include effective denoising techniques and uncertainty assessment to address the practical problems in wearable or emergency contexts.

### 1) FUTURE WORK

To further enhance the clinical usefulness and strength of the proposed ECG-CNNTX architecture, future work will mainly concentrate on cross-dataset validation utilizing many independent and different ECG datasets, such as PTB-XL, INCART, and AFDB. This stage is important to test how well the model can work with other types of patients, settings, and annotation standards, so that it can be trusted to work with more than just the MIT-BIH dataset.

The framework will also be expanded to include multi-lead ECG data in order to capture the inter-lead relationships that are necessary for a complete cardiac diagnosis in clinical situations. The current binary classification (PVC vs. Normal) scheme will be expanded into a multi-class prediction framework to detect a broader spectrum of arrhythmias, enabling the model to handle clinically relevant rhythm abnormalities such as atrial fibrillation, bundle branch block, and supraventricular tachycardia.

To enable real-time prediction and edge deployment, the future study will investigate model compression and optimization methodologies, including pruning, quantization, and knowledge distillation, aimed at diminishing computing complexity while preserving elevated diagnostic precision. These improvements will make it possible to use the software on embedded and wearable devices with limited resources, which will allow for continuous heart monitoring and point-of-care diagnostics. Lastly, lightweight deployment frameworks like TensorFlow Lite and ONNX Runtime will be used to test the model's latency, memory use, and energy efficiency. This will show that it can be used in telehealth and clinical monitoring settings in real time.

Future improvements may include the implementation of attention rollouts for enhanced interpretability, the use of knowledge distillation for model compression, and the integration of temporal information (e.g., patient history or heart rate variability) for contextually aware predictions. These developments would enhance the therapeutic use of AI in cardiovascular care and facilitate the model's practical implementation.

## V. CONCLUSION

The proposed ECG-CNNTX hybrid model effectively integrates convolutional and transformer-based architectures to identify ECG data converted into image format. Through rigorous testing and hyperparameter optimization, the model attained a stable validation accuracy of 99% with reduced training loss, while exhibiting significant computing efficiency. The model's proficiency in generalization, rapid convergence, and precise differentiation between PVC and regular beats render it appropriate for practical use in early-stage arrhythmia detection systems. Moreover, the incorporation of Grad-CAM-based visual explanations enhances trust and transparency in clinical applications, enabling medical professionals to discern the areas that affect the model's decisions. Experimental findings validate that our approach not only attains state-of-the-art accuracy but also enhances the reliability of deep learning-based ECG classification systems.

The proposed technique demonstrates robust efficacy in arrhythmia detection; however, other potential pathways remain for further investigation. The existing binary classification framework that differentiates PVC from non-PVC beats may be expanded to a multi-class model, with real-time implementation on a wearable ECG device, and validated on diverse datasets. Moreover, it is essential to investigate model compression techniques, such as pruning, quantization, and knowledge distillation, to enable efficient inference in low-power edge environments. Ultimately, enhancing the model's explainability via the incorporation of advanced interpretability tools like SHAP, LIME, attention rollouts, and Grad-CAM may provide significant insights into the model's decision-making process and improve its dependability in clinical applications.

## DECLARATION OF COMPETING INTEREST

The authors declare no conflict of interest.

## DATA AVAILABILITY

The data used in this study are from publicly available datasets. The datasets used are as follows: <https://physionet.org/content/afdb/1.0.0/>.

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